

Companion LXXIII: Maximum Mean Discrepancy of Persistence Diagrams in the Relaxation-Driven Cyclic Cosmology

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Abstract

The maximum mean discrepancy (MMD) provides a kernel-based statistical test for distinguishing probability distributions in a reproducing kernel Hilbert space (RKHS). Applied to persistence diagrams, MMD quantifies whether two ensembles of topological features arise from the same underlying cosmological model. In weak-lensing analyses, this enables a direct comparison between the Relaxation-Driven Cyclic Cosmology (RDCC) and the standard Λ CDM model. Building on the Hilbert-space embeddings and kernel PCA developed in previous Companions, this work derives the MMD for persistence diagrams in RDCC, identifies the scale-dependent signatures of potential relaxation, and establishes a statistically rigorous framework for model discrimination using topological data analysis. The resulting test provides a robust, nonparametric, and observationally accessible probe of the RDCC gravitational field.

1 Introduction

Persistence diagrams encode the birth and death of topological features in weak-lensing convergence fields. Hilbert-space embeddings and kernel PCA provide a linearized and spectrally decomposed representation of these diagrams. Maximum mean discrepancy (MMD) extends this framework by enabling statistical tests between two distributions of embedded diagrams.

In cosmology, MMD offers a powerful tool for distinguishing between models based on their topological signatures. In RDCC, the relaxation mechanism modifies the scale-dependent evolution of the gravitational potential, altering the distribution of persistence diagrams in a characteristic way. MMD provides a direct, nonparametric test for detecting these modifications in weak-lensing surveys.

2 Maximum Mean Discrepancy for Persistence Diagrams

Given two ensembles of persistence diagrams

$$\{D_i\}_{i=1}^n, \quad \{E_j\}_{j=1}^m,$$

their embeddings into an RKHS $\mathcal{H}$ are

$$\Phi(D_i), \quad \Phi(E_j).$$

The squared MMD is defined as

$$\text{MMD}^2 = \left\| \frac{1}{n} \sum_{i=1}^n \Phi(D_i) - \frac{1}{m} \sum_{j=1}^m \Phi(E_j) \right\|_{\mathcal{H}}^2 \quad (1)$$

$$= \frac{1}{n^2} \sum_{i,i'} k(D_i, D_{i'}) + \frac{1}{m^2} \sum_{j,j'} k(E_j, E_{j'}) - \frac{2}{nm} \sum_{i,j} k(D_i, E_j). \quad (2)$$

This quantity is zero if and only if the two distributions are identical.

3 RDCC Modification of Persistence Diagram Distributions

In RDCC, the convergence evolves as

$$\kappa_{\text{RDCC}}(\ell) = \kappa(\ell) \left[1 - \chi_{\kappa} \left(\frac{\ell}{\ell_{\text{rel}}} \right)^{\alpha} \right]. \quad (3)$$

This modifies the birth and death thresholds of topological features, shifting the distribution of persistence diagrams:

$$D \mapsto D_{\text{RDCC}}.$$

The kernel becomes

$$k_{\text{RDCC}}(D_i, D_j) = k(D_i, D_j) \left[1 - \chi_k \left(\frac{\ell}{\ell_{\text{rel}}} \right)^{\alpha} \right].$$

4 MMD in RDCC

The RDCC-modified MMD is

$$\text{MMD}_{\text{RDCC}}^2 = \text{MMD}^2 \left[1 - \chi_{\text{MMD}} \left(\frac{\ell}{\ell_{\text{rel}}} \right)^{\alpha} \right]. \quad (4)$$

RDCC therefore induces:

- increased separation from ΛCDM at large scales,
- suppressed separation at small scales,
- a characteristic turnover near ℓ_{rel} ,
- enhanced sensitivity to long-lived persistence features,
- modified kernel expectations across redshift.

5 Hypothesis Testing: RDCC vs. ΛCDM

Define the null hypothesis

$$H_0 : \mathcal{P}_{\text{RDCC}} = \mathcal{P}_{\Lambda\text{CDM}},$$

and the alternative

$$H_1 : \mathcal{P}_{\text{RDCC}} \neq \mathcal{P}_{\Lambda\text{CDM}}.$$

The test statistic is $\text{MMD}_{\text{RDCC}}^2$.

Permutation tests or asymptotic bounds yield p -values.

RDCC predicts:

- significant rejection of H_0 at large scales,
- marginal separation at small scales,
- maximal discriminative power near ℓ_{rel} .

6 Conclusion

Maximum mean discrepancy provides a statistically rigorous method for distinguishing the Relaxation-Driven Cyclic Cosmology from Λ CDM using persistence diagrams. The relaxation mechanism modifies the distribution of topological features in a scale-dependent manner, producing distinctive signatures in the MMD test statistic. These effects offer a direct, nonparametric, and observationally accessible probe of the RDCC gravitational field. The present Companion establishes the foundation for future studies of statistical topology, model discrimination, and cosmological inference in RDCC.

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